

SIMATS ENGINEERINGASSESSMENT TOOL 3
COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

NAME:K.VIJAYALAKSHMI

REG.NO:192511189

Q1: Smart Banking System

- (a) Class: blueprint
Object: instance
Message Passing: communication between objects
Encapsulation: data hiding
Polymorphism: multiple forms of methods.
- (b) Association: general relationship
Aggregation: weak whole-part
Composition: strong whole-part.
- (c) Improves reusability via modular classes and maintainability via encapsulation and clear relationships.

Q2: Smart Healthcare System

- (a) Abstraction: hiding complexity;
Encapsulation: bundling data;
Modularity: dividing system.
- (b) LSP: subclasses replace base classes
ISP: small specific interfaces.
- (c) Improves flexibility, reduces coupling, increases quality.

Q3: SDLC

- (a) Phases: Planning, Requirement Analysis, Design, Development, Testing, Maintenance.
- (b) Each ensures structured development and error reduction.
- (c) Leads to reliable and high-quality software.

Q4: Agriculture System

- (a) Prototype, Spiral, Agile, DevOps explained.
- (b) Prototype vs Agile: Prototype is early model; Agile is iterative delivery.
- (c) Agile is most suitable due to changing requirements.

Q5: UML

- (a) UML helps visualize design.
- (b) State, Component, Deployment, Package diagrams.

(c) Improves stakeholder communication.

Q6: Inventory System

(a) Modular design divides system.

(b) Procedural vs OOP: step-based vs object-based.

(c) Improves scalability, maintainability, reusability.

